

GL-ProQ-en-014

Supplier Production Part Approval Process (PPAP OG)

Process Owner: MICKAEL SICAIRE

Approver 1: CHARLES JERVAIS

Approver QMS: ERIC JACQUOT

Revision: 3.0

Effective Date: 08/04/2025

Revision Date: 08/04/2025

Life Is On



Purpose

The purpose of this procedure is to define Supplier Production Part Approval Process (PPAP) as an output of the Supplier Advanced Product Quality Planning (APQP) process to qualify Purchased parts and products pertaining to manufacturing and assembly.

Scope

This procedure shall be applicable to:

- All production Bought Out Parts (BOP)/products, Subcontracted parts/products.
- The below situations in which Schneider Electric needs to manage any part/product change(s):
 - Change of Supplier (CoS), Power of 2(Po2), and Part/product localization.
 - Tooling Transfer, Refurbishment (Including Change of cavity, major repair and not applicable to minor and preventive repairs, as per Lending for Use Agreement (LFU)), Brand-new tooling or Duplication.
 - Supplier Process Change Notification (SPCN).
 - New Design & Quality & Value Engineering (QVE).
 - All Category specified by Schneider Electric such as Mechanical, Electrical & Electronics (EE), Electronic Manufacturing Services (EMS), Fabricated Component (FC), Brand Labelling (BL) offer...

This procedure is applicable for all Entities of Schneider Electric: GSC, Global ETO, LOB's & Manufacturing.

Target audience

All Relevant functions but in particular:

Global Supply Chain (GSC)

- Procurement & Supplier Quality and Development
- Customer Satisfaction and Quality Teams (CS&Q)
- Industrial Teams

Business Units and Lines of Businesses (LOB's)

- Engineering Design Teams
- Industrialization teams
- Customer Satisfaction & Quality Team

Table of Contents

Topic	See Page
Contents	
References.....	2
Process Inputs	2
Process Overview.....	3
PPAP Deliverables.....	4
PPAP Target.....	4
PPAP Submission Level.....	5
Part/Product Qualification Flow Chart	6
Significant production run	6
Digital tools	6
Process Outputs.....	6
Monitoring & Control.....	7
Roles & Responsibilities	7
Competency Requirements	7
Quality Records	7
Terms & Acronyms.....	8
Appendix A: Process Map	9
Appendix B: CoS & Localization, Tooling Transfer, New Tool, SPCN, Raw	
Material Changes	10
Appendix C: Common parts	11

References

- **GL-DirQ-050 Schneider Electric Supplier Quality Management Directive**
- **PPAP BOOK**
- **Supplier Quality Excellence Guide**
- **GL-ProQ-037 Supplier Advanced Product Quality Planning (APQP)**
- **OLM Offer Lifecycle Management & OCM Offer Change Management**
- **GL-DirQ-032 Directive Plants-Lines of Business quality execution**
- **GL-ProQ-022 CTQ/CTP Management Procedure**
- **GL-ProQ-017 Process Failure Modes & Effects Analysis (PFMEA) Procedure**
- **GL-ProQ-034 Control Plan Procedure**
- **GL-ProQ-033 Conduct Process Capability Study and Analyse Results Procedure**
- **Internal Schneider Electric Encyclopedia iSEE**

Process Inputs

- **SOC - Supplier Orientation Chart**
- Design for Safety & Reliability (DfSR) data - Critical to Quality & Safety Specifications (CTQ & CTQ S&R)
- Part/Product(s) specifications (Supplier Package from PDM)
- Supplier APQP (Advanced Product Quality Planning) data
- iSave and PCR for Legacy Offer

**Process
Overview**

The Supplier PPAP process shall be applicable for any new Parts/Products, part modifications, or parts produced from new or revised production and/or assembly processes. Any change(s) shall be covered by a new Supplier PPAP validation as to be defined and communicated by Supplier Quality Representative.

Supplier PPAP is one of the last deliverables of the Supplier APQP process. The results of the Supplier PPAP submission provides Schneider Electric all reasonable evidence that our Suppliers have met the customer defined requirements (including Safety and Regulatory requirements), and the manufacturing and assembly process shall be capable of consistently producing quality parts at the agreed-on production rate.

For any catalogue parts/products not requiring customization, no additional Supplier PPAP requirements are needed. For catalogue parts/products with any type of customization or any changes required to Supplier Data Sheet, a full PPAP shall be required.

If any Product/Part specifications cannot be met, Supplier shall document their problem-solving efforts and contact Schneider Electric Quality Representative for concurrence in determination of appropriate corrective action.

The Supplier PPAP includes: (see **process map in [Appendix A](#)**)

- **[PPAP Deliverables](#)**
 - **[PPAP Target](#)**
 - **[PPAP Submission Level](#)**
 - **[Part/Product Qualification Flow Chart](#)**
 - **[Significant Production Run](#)**
-

**PPAP
Deliverables**

Supplier PPAP Acceptance Level shall be applicable for all production parts/products intended to be used on our business.

Supplier PPAP Acceptance Level shall be provided to supplier by Schneider Electric Supplier Quality Representative under any of the following outcomes:

- **Fully approved**
 - All products/parts, materials, and/or related documentation (or records submitted to, and or reviewed by Supplier Quality Representative) shall be approved only when they meet all Schneider Electric requirements. After this approval is granted, the supplier is authorized to ship out the respective parts and/or products.
 - Supplier PPAP book and PSW (Supplier PPAP approval) needs to be stored in **PDM**.
 - If there are any gaps/deficiencies identified in the Supplier PPAP book, the Supplier PPAP cannot be approved.
- **Conditional or Interim Approval**
 - Shall be granted when the conditional or defined interim Supplier PPAP targets have not been achieved but the risks remaining have been sufficiently mitigated for the derogation period (Temporary Quality Wall, Final Inspection). These are to be time-based not more than 6 months to close all remaining actions successfully.
- **Rejected**
 - Supplier PPAP shall be rejected if submission does not meet Schneider Electric requirements, based on the production lot from which it was taken and/or accompanying documentation. In such cases, the submission and/or process, as appropriate, shall be corrected to meet Schneider Electric requirements. The submission shall be approved before production quantities may be shipped.

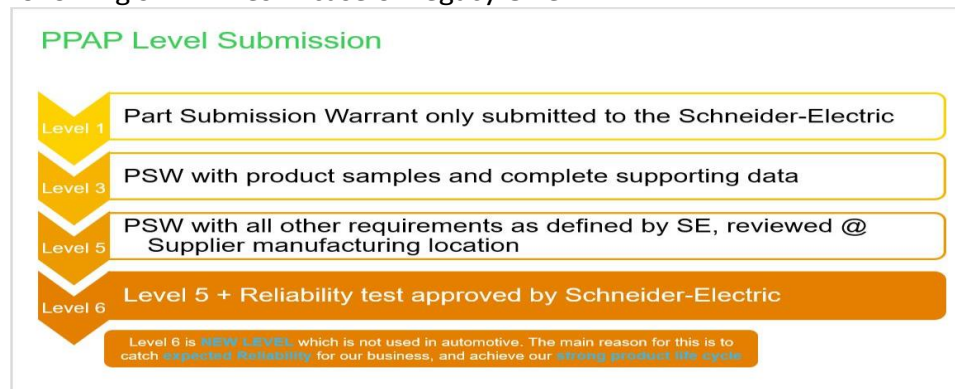
PPAP Target

To achieve full Supplier PPAP Approval, the following steps shall be successfully completed, otherwise proper deviation shall be put in place, in which case, Supplier PPAP status will be considered as Conditional or Interim Approval with Deviation:

- Part/product 100% compliance with requirement, design record specifications: Material, Dimensional, Performance...
- Capability Indexes achieved for Critical to Quality Safety & Regulation (CTQ S&R) or key characteristics.
- Process validation & part/products compliance with all CTQ S/R.
- PPAP approved on Time.

PPAP Submission Level

Supplier PPAP Level to be determined from Activity 6: Risk Assessments conducted as Stage 0 of Supplier APQP process for New Offer, or by following swim lines in case of Legacy Offer.



For New Offer: ASQ must approved all Supplier PPAP Levels before transition to GSC or Global ETO organization.

For Legacy Offer: All Supplier PPAP Level 1 and Level 3 must be approved by GSC or Global ETO (all PPAP iTems under Schneider-Electric responsibility or support & PSW), LOB must be involved if Supplier/GSC identify any gaps between samples and part prints.

All PPAP Level 5 and Level 6, LoB team members must be involved. (For approval definition, you must refer to [Deliverables chapter](#))

As per the Supplier PPAP Level, documents to be Submitted or Retained by supplier are the following:

PPAP Deliverables Vs Submission Levels

Schneider	Supplier	Level 1	Level 3	Level 5	Level 6	PPAP Requirement	Responsible	Who can support	APQP Start	APQP End
		S	S	S	S	1- Design record	Schneider		0-3	I-16
		*** SSP-SRM → SPCN ***				2- Supplier Engineering change documents	Supplier / SQ		0-2	
		S	S	S	S	3- Schneider-Electric Engineering approval (if required)	Supplier / Design		0-4	
		R	R	S	S	4- Design FMEA (if required)	Supplier / Design		0-3	I-12
		R	R	S	S	5- PFD-Process Flow Diagram/PFC-Process Flow Chart	Supplier		I-11	II-17
		R	R	S	S	6- Process FMEA	Supplier		I-11	II-18
		R	R	S	S	7- Supplier Quality Control Plan	Supplier		I-11	II-20
		R	R	S	S	8- MSA-Measurement System Analysis	Supplier		II-19	III-23
		S	S	S	S	9- Dimensional Result, FAIR	Supplier / Design		I-14	III-23
		S	S	S	S	10- Material Performance results and Test reports	Supplier / Design		I-14	III-23
		R	R	S	S	11- Initial Process Studies – STPC	Supplier / Design		I-14	III-25
		R	S	S	S	12- Qualified Laboratory Documentation	Supplier		II-19	III-23
		S	S	S	S	13- AAR-Appearence Approval Report (if applicable)	Supplier / Design		I-14	III-23
		R	S	S	S	14- Sample Product	Supplier		I-14	III-24
		R	R	R	R	15- Master Sample	Supplier			III-24
		R	R	R	R	16- Checking aids	Supplier		II-17	II-22
		R	R	S	S	17- Records of Compliance	Supplier		I-11	III-24
		NA	NA	NA	S	18- "New" Product Reliability Testing	Schneider		I-15	IV-29
		S	S	S	S	19- PSW-Part Submission Warrant	Supplier / SQ		IV-29	IV-29

Reliability Focus

S = Submitted R = Retained

Life Is On Schneider Electric

Confidential

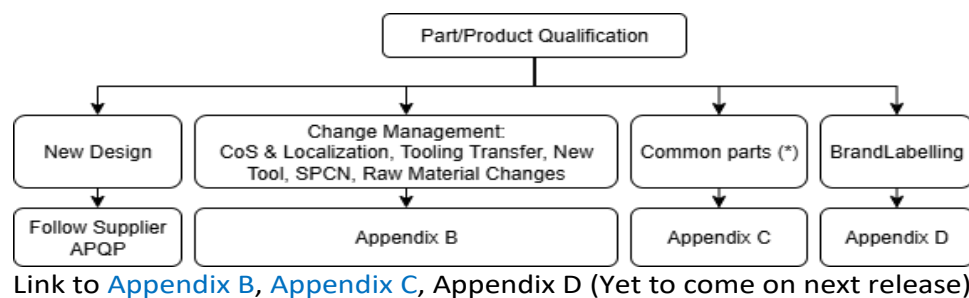
Retained

- The supplier shall retain at appropriate location and make available to Schneider Electric upon request.

Submitted

- Supplier shall submit all documentation to Schneider Electric and retain a copy for their own records. Table shown above shows all the deliverables for each level of the Supplier PPAP defined.

**Part/Product
Qualification
Flow Chart**



(*) Common Part:

From an established flow between Local "supplier & plant" in same location.

1. Same Design already used from PDM.
2. If any approved deviation exists, it must be accepted as is for the new flow under creation.
3. Approved by PPEP/PPAP
4. Same supplier processes to manufacture the parts.
5. Packaging used in current flow can be adjusted for the exported flow.

**Significant
production run**

For Supplier PPAP production, Parts, Sub-Assemblies, or product shall be taken from a significant production run. This significant production run shall be from 1 Hour to 8 Hours of production, and with the specific production quantity to total a minimum of 300 consecutive parts, unless otherwise specified by Schneider Electric representative.

This significant production run shall be conducted at the production site, at the agreed-on production rate, using the production tooling, production gauges, production process, production materials, and production/assembly line operators.

Parts from each unique production process, each position of a multiple cavity die, mold, tool, or pattern, shall be measured and representative parts tested.

Digital tools

- [APQP/PPAP SharePoint](#)
- [PPAP BOOK](#)
- [PPAP UI from SSP-SRM](#)
- [Jira Global Supplier APQP-PPAP \(GSAP\)](#)

**Process
Outputs**

- Part Submission Warrant (PSW) Signed
- Product/Part Qualified
- Supplier PPAP documents package store in PDM
- PVP Product Verification Plan if required
- Purchase Order

Monitoring & Control

The audit program for monitoring/evaluation reference **GL-DirQ-009 Internal Audit Directive**

Roles & Responsibilities

See swim lines in appendixes [A](#), [B](#), [C](#)

Competency Requirements

Refer to [SoR](#) (System of Reference: Job Codes & Competencies), [LoP](#) (Levels of Proficiency) & [CoMET](#) for detailed competencies.

FQ06 → Supplier Qualification and Development: Ability to audit existing and potential suppliers to determine their capability to deliver required good on time and coach the suppliers to match Schneider's expectations.

FQ06-2 → (Level 3) Auditing and Compliance: Planning, Conducting, Reporting and Closing findings related to the effectiveness of supplier audits in the following categories: Supplier QMS, process audits, product audits, technical audits, compliance audits (and confidential investigations as required). Auditing includes the ability to track, report, close and escalate for systemic breakdowns. Additional risk management is required for, regulatory, HSE.

FQ06-7 → (Level 5) PPAP (Production Part Approval Process) - part qualification: Understands PPAP as an outcome of effective APQP. Excellent judgement in PPAP level requirements and the 18 deliverables for evaluation and approval to the supplier. Ability to effectively evaluate PPAP submissions and hold suppliers accountable to process run capability and traceability requirements.

FQ06-9 → (Level 5) Core Tools (PFMEA, Control Plan, MSA, SPC, Poka-Yoke, GD&T, Blueprint Reading...: Deep evaluation and application knowledge of all core tools required for entire SQ community. AIAG standards apply for each category except GD&T and Blueprint Reading. For GD&T & BPR, the ability to read and interpret prints and standards including basic tolerancing and advanced tolerancing. Ability to design gages and fixtures based on dimensioning and tolerancing standards.

Quality Records

Record	Record Location	Retention Period
PSW	PDM	Life of product + 5 years
PPAP BOOK	PDM	Life of product + 5 years

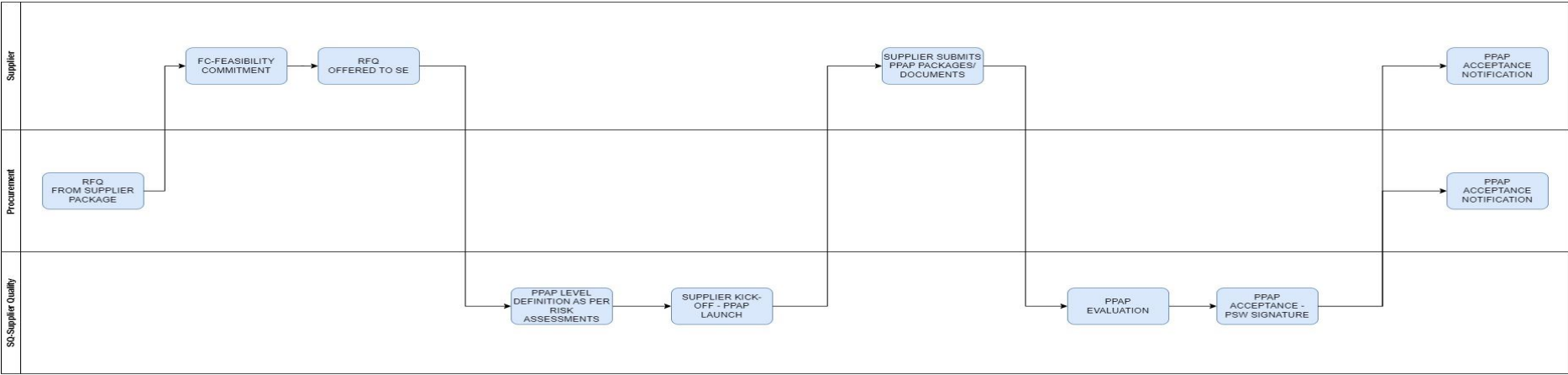
**Terms &
Acronyms**

The table below includes definitions for terms and acronyms used in this document. Reference [iSEE](#) for a full list of terms and acronyms used in the SE IMS.

Term	Definition
3F	Fit, Form & Function
APQP	Advanced Product Quality Planning
ASQ	Advanced Supplier Quality
BL	Brand Labelling
COS	Change Of Supplier
CQCS	Component Quality Control Specification
CTP	Critical To Process
CTQ	Critical To Quality
CVP	Component Validation Plan
DFSR	Design for Safety and Reliability
EE	Electrical & Electronics
FAIR	First Article Inspection Report
FC	Fabricated Components
GSC	Global Supply Chain
LAIR	Last/Light Article Inspection Report
LFU	Lease for Usage Agreement
LOB	Line of Business
LTC	Logistic Terms & Conditions
MCSV	Mechanical Component and Subassembly Verification
MSA	Measurement System Analysis
OLM	Offer Life Management
PDM	Product Data Management
Po2	Power of Two
PO	Purchase Order
PPAP	Production Part Approval Process
PPEP	Production Part Evaluation Process
PRB	Problem
PSW	Part Submission Warrant
PVP	Product Validation Pla
QVE	Quality & Value Engineering
RA	Risk Assessment
RPN	Risk Priority Number
SAM	Supplier Approval Module
SOD	Safety Occurrence Detection
SPCN	Supplier Process Change Notification
SSP-SRM	SE Supplier Portal - Supplier Relationship Management
SSQM	Schneider Supplier Quality Management
STPC	Short Term Process Capability
S&R	Safety & Regulation
TA	Technical Antenna
TR	Tolerance Range
VF	Virtual Factory
VP	Validation Plan

Supplier Production Part Approval Process (PPAP OG)

Appendix A: Process Map



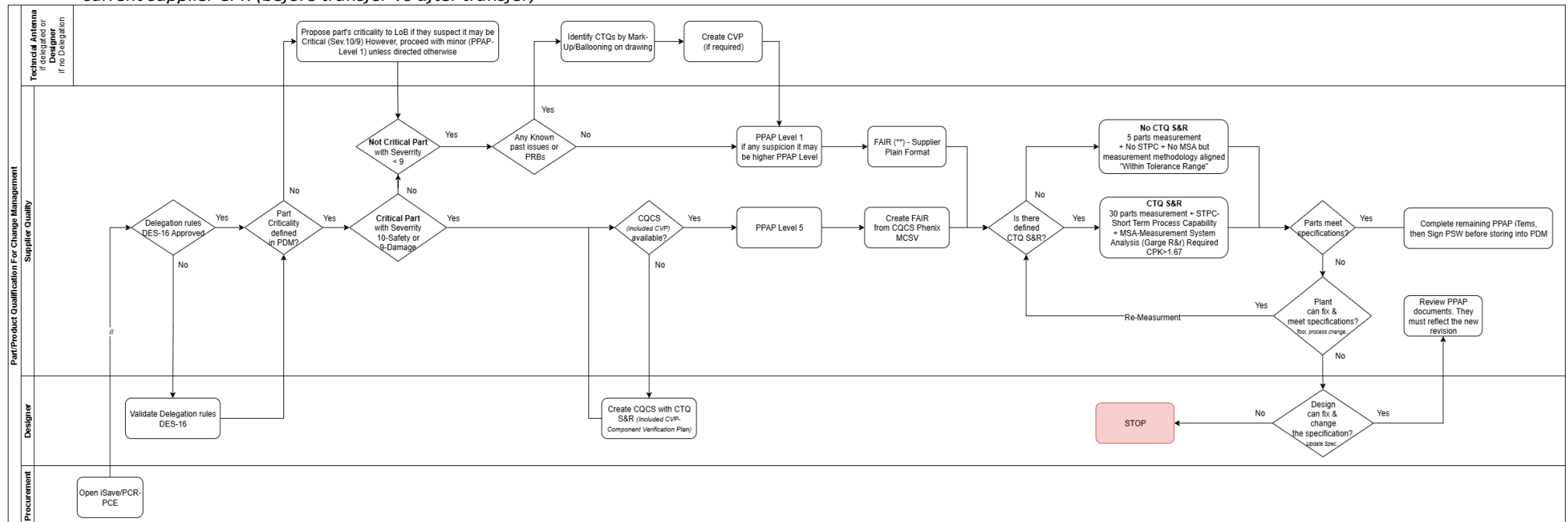
General

Supplier Production Part Approval Process (PPAP OG)

Appendix B: CoS & Localization, Tooling Transfer, New Tool, SPCN, Raw Material Changes

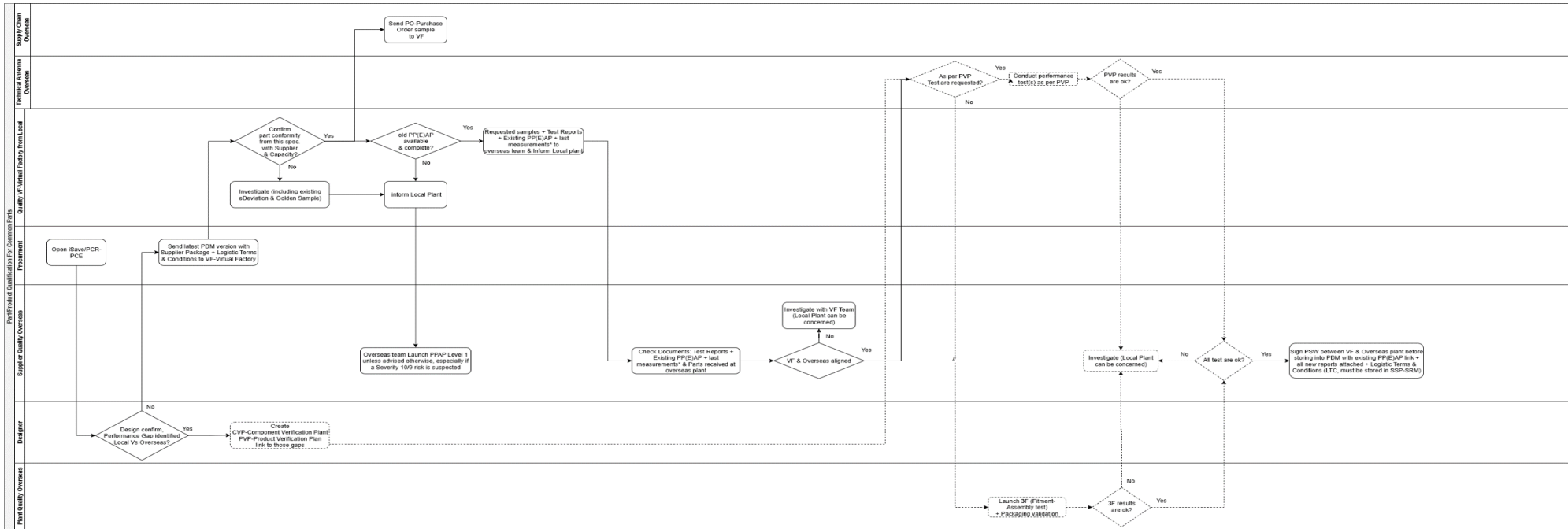
Delegation rules DES-16 Link: https://schneiderelectric.sharepoint.com/:p:/r/sites/olm-library/_layouts/15/Doc.aspx?sourcedoc=%7Bd7776803-6855-427b-8469-1b3fd40a111f%7D&action=edit&mobileredirect=true

() Exit PPAP or at least LAIR & FAIR (LAIR-Last/Light Article Inspection Report) must be launched in case of CoS-Change of Supplier & Tool Transfer. New CPK must be >= current supplier CPK (before transfer Vs after transfer)**



General

Appendix C: Common parts



The Overseas Team or Plant refers to a new team or Plant that intends to reuse a part originally developed and qualified by the local team.

(*) Last measurements: The latest local plant or supplier batch inspection results (≤ 3 months) to confirm PPAP compliance. If > 3 months, submit a LAIR (selected list of characteristics, including at least CTQs) manage by VF. In any case, measurement methodology must be aligned between Local Plant, Supplier Overseas plant

Dotted line: Not under the responsibility of Supplier Quality Team

General

REVISION HISTORY SUMMARY

Approvers	Requested By	Change Description	Reason for Change	Revision	Revision Date	Contributors
ERIC JACQUOT; CHARLES JERVAIS	MICKAEL SICAIRE	1. Adapt the specification for Legacy Offer, to accelerate our Part/Product Qualification and support our business 2. Major evolution on the appendix B & C in the swimlines	Simplification the Part/product Qualification from Supplier	3.0	08/04/2025	
MAGDALENA MICHULEC; CHARLES JERVAIS	MICKAEL SICAIRE	* This document has been migrated from the Global QMS Documents Platform (SharePoint) without revision changes. Process owner/Document author and revision details are specified in the document itself. * A summary of previous revisions is available only in the migrated version.	Document migration	2.1	12/10/2024	